CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 6: **How Forces Affect Motion** | Quick revision, one page

Newton's laws

- First law (law of inertia): a body remains at rest or in uniform motion in a straight line unless an external unbalanced force acts on it. Inertia depends on mass.
- Second law: the rate of change of momentum is proportional to the applied force, in the direction of the force: $F = \dfrac{\Delta p}{\Delta t} = ma$.
- Third law: to every action there is an equal and opposite reaction; the two forces act on *different* bodies.

Force, momentum and weight

- SI unit of force: newton, $1\ \text{N} = 1\ \text{kg m/s}^2$.
- Momentum $p = mv$ (kg m/s), a vector in the direction of velocity.
- Weight $W = mg$ (in N); $g \approx 10\ \text{m/s}^2$ on Earth, about one-sixth of this on the Moon; mass is constant, weight varies with g.

Conservation of momentum

- In an isolated system, total momentum before = total momentum after: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.
- Recoil of a gun (both initially at rest): $0 = mv + MV$, so $V = -\dfrac{mv}{M}$.
- Rocket: hot gases ejected backward give the rocket momentum forward.

Friction

- Friction opposes the relative motion (or tendency of motion) between surfaces in contact; it depends on the nature of surfaces and the normal force.
- Rolling friction $<$ sliding friction; static friction $\geq$ sliding friction.
- Reduced by lubricants, polishing, ball bearings, wheels. Useful for walking, gripping, braking.

Applications of $F = ma$

- Net force $F_{\text{net}} = F_{\text{applied}} - f$ when friction f opposes the motion.
- Impulse $= F\Delta t = \Delta p$; increasing the time of impact reduces the force (cricket catch, air bags).

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